

Lost Star: An Interactive Stereoscopic Picture Book Installation for Children's Bedtime Rituals

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Abstract

Sleep disorders in early childhood hinder development and disrupt daily functions, but effective bedtime rituals can help. However, the integration of tangible and virtual interactions in sleep-related design projects remains unexplored. This paper investigates how pop-up books and multimedia can create a playful and accessible way for children to engage in bedtime activities. We present the design of Lost Star, a sonic and tactile Tangible User Interface (TUI) designed to enhance the quality of bedtime rituals for children with sleep problems. Additionally, we explore Joint Media Engagement (JME) to foster meaningful interactions between parents and children, enhancing bedtime routines and improving sleep outcomes.

CCS Concepts

• **Human-centered computing** → **Tangible UI, interaction design.**

Keywords

Tangible interaction, three-dimensional picture books, sleep disorders, sleep rituals

ACM Reference Format:

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1 Introduction

Child sleep problems, like night waking and difficulty falling asleep, impact development and parental well-being [8]. Strategies like staying with the child or letting them cry provide short-term relief but don't reduce long-term parental fatigue. A solution is needed to improve children's sleep quality and reduce parental stress. Stable bedtime rituals enhance sleep and parental happiness. Picture books help children fall asleep faster, improve sleep quality, and foster parent-child relationships [6].

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We propose an interactive picture book device for bedtime rituals, combining tangible and virtual elements. It automates page-turning, offers interactive content, and provides soothing storylines to enhance companionship. Multi-sensory experiences, such as hypnotic music and tactile stimulation, will engage children and aid sleep. The device stimulates imagination and creativity with audio-visual effects, improving sleep quality and fostering reading habits. Joint Media Engagement (JME) will enhance parent-child interactions and bedtime rituals. Our installation aims to improve children's sleep, help them transition to sleeping alone, and promote overall health, giving parents more personal space and reducing anxiety.

2 Related Work

Traditional bedtime routines, such as reading and singing lullabies, improve children's sleep quality and mitigate behavioral sleep disorders. Structured bedtime activities reduce bedtime resistance and nighttime awakenings in children [1]. Behavioral interventions, like positive reinforcement, are beneficial for treating pediatric sleep disorders. Interactive multimedia tools are popular for sleep rituals, especially when integrated with picture books. These books combine storytelling with sensory stimuli to engage children and make bedtime more enjoyable [2]. Interactive stories with audio-visual elements create a calming environment for children, reducing anxiety and aiding quicker sleep onset [7]. Combining picture books with multimedia elements like animations and sounds helps address children's sleep difficulties [4]. The American Academy of Pediatrics recommends using JME to improve children's sleep quality and overall health [3]. JME enhances children's learning and emotional bonds, promoting mutual engagement and dialogic inquiry, improving bedtime routines. Tseng et al. designed PlushPal to combine physical toys with digital content to enhance bedtime activities [5].

3 LOST STAR: Stereoscopic Interactive Picture Book System

3.1 Overview of Lost Star

Lost Star is a three-dimensional interactive picture book system that combines tangible and virtual elements designed to help children ages 3-6 with sleep disorders establish a coherent, engaging bedtime ritual through bedtime parent-child interaction. Our device consists of a physical flat part of the picture book, a three-dimensional part of the picture book, a projector, and a laptop running the application (see Figure 2).

Lost Star has three main design features: (1) *Three-dimensional Book Physical Interaction Form*: The three-dimensional picture book serves as the primary medium for the bedtime ritual. Through the

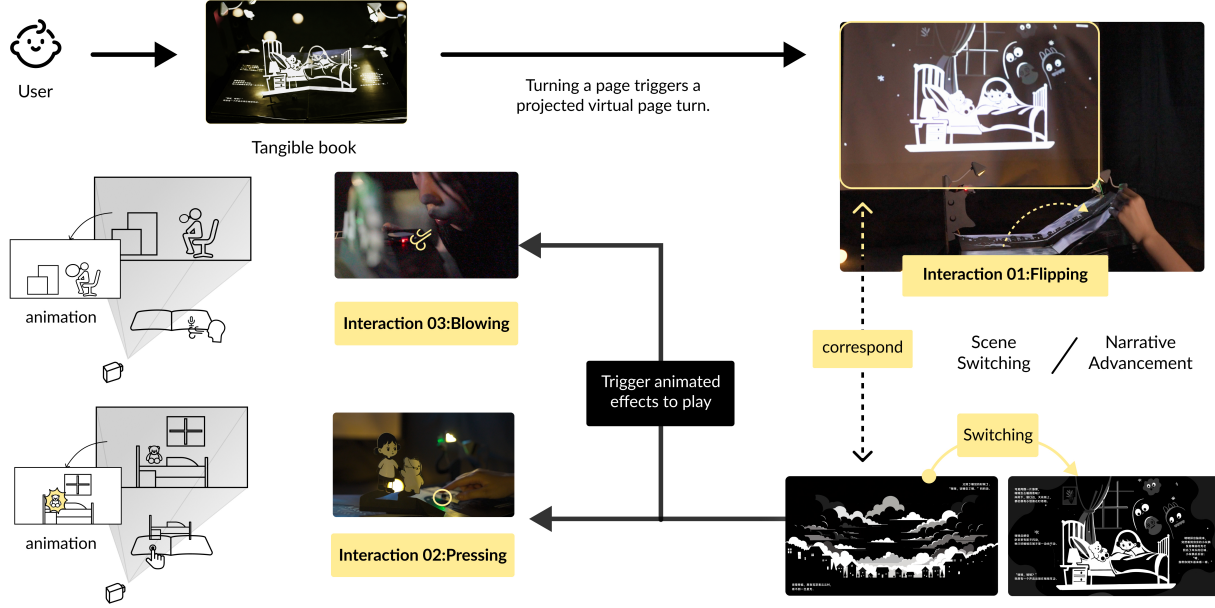


Figure 1: Schematic diagram of the interaction of a physical interactive book and the related interaction flow description.

engaging format of the three-dimensional book and simple interactive methods, we focus the child’s attention on the picture book story. The format of the three-dimensional picture book not only provides visual and tactile experiences but also supports parent-child interaction, enhancing overall well-being. (2) *Multi-sensory Experience and Dynamic Audiovisual Feedback*: The picture book acts as an interactive input device. To enhance the multi-sensory experience, we incorporate projection effects as dynamic audiovisual feedback, advancing the story’s plot while simultaneously encouraging and supporting the child’s cognitive development. (3) *Integration of Multimedia Devices in Bedroom Environments*: Lost Stars offers the possibility of using multimedia devices in the bedroom environment. The system supports interactive parent-child reading while also adapting to automated reading for the child, offering high flexibility and usability (see Figure 1).

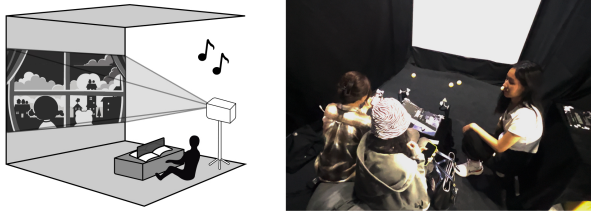


Figure 2: Installation layout and site situation.

3.2 Implementation

In the overall system implementation, we divided our project into physical and virtual components. For the physical part, we hand-crafted the pages of the pop-up book and used laser-cutting technology to create intricate cut-outs, along with manually crafting the

pop-up mechanisms. In the projection component, we designed and developed the virtual aspects of the picture book using the Unity system, integrating Arduino to control sensors that connect seamlessly to Unity. We utilized MadMapper to adjust the resolution and range of the projector (XGIMI H3S). The picture book is equipped with three button sensors and an airflow sensor to meet the diverse interaction needs of children and parents. As children read the book, they can discover the connection between pressing buttons and multimedia interactions, allowing them to explore the story independently. To ensure completeness and a soothing bedtime experience, we commissioned professional voice recording and selected background music designed to promote sleep, featuring soft, simple harmonic structures and high repetition of instrumental music.

4 Conclusion

As we explore the issue of sleep disorders in children, better integrating JME into the form of three-dimensional books has become a key breakthrough in our interactive approach. We introduce Lost Star, designed to help children aged 3-6 with sleep disorders establish a consistent and engaging bedtime ritual. This innovative approach combines tangible interaction with virtual elements. Initial tests show positive feedback from child development experts, who believe it can create a new type of bedtime ritual. Future plans include expanding the system to families of children with sleep disorders and enhancing its universal applicability.

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